

ECE 3337: Signals & Systems Analysis

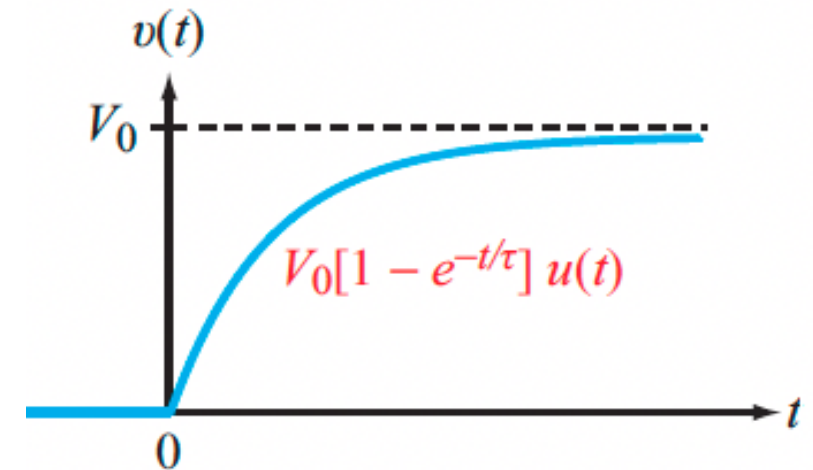
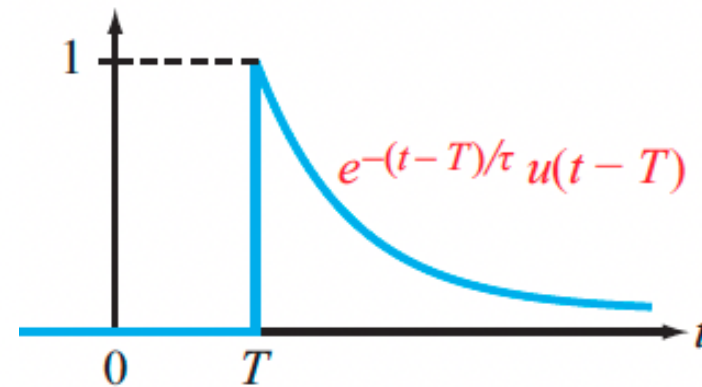
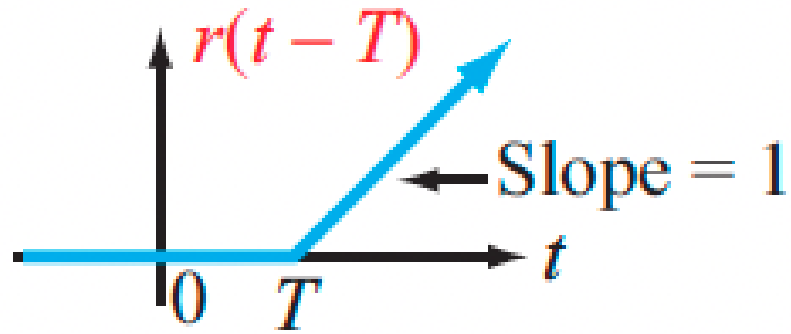
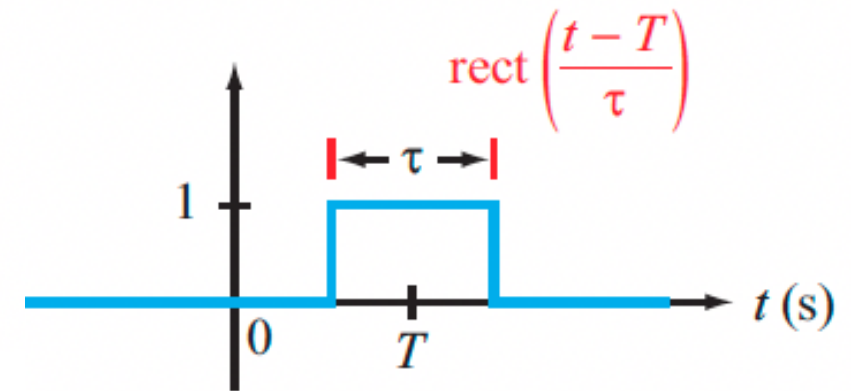
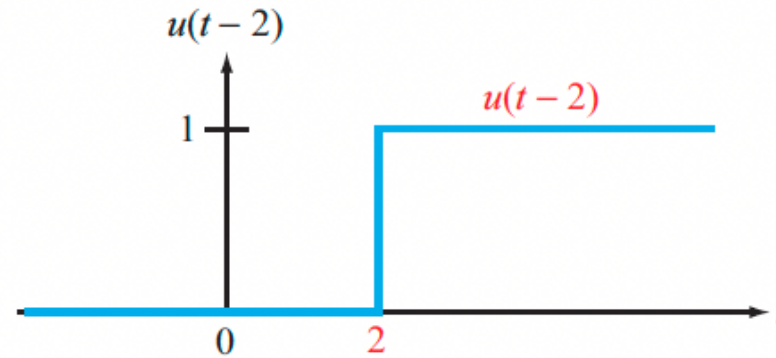
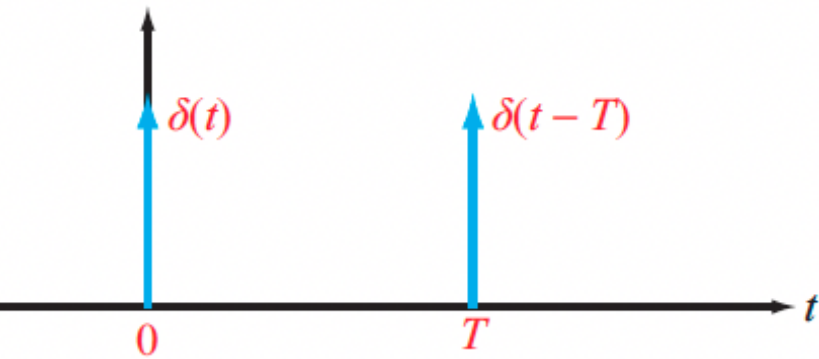
Lecture 4: Linear Time Invariant Systems

**Reference: Chapter 2
(Sections 2.1 – 2.2)**

Please turn off your gadgets now



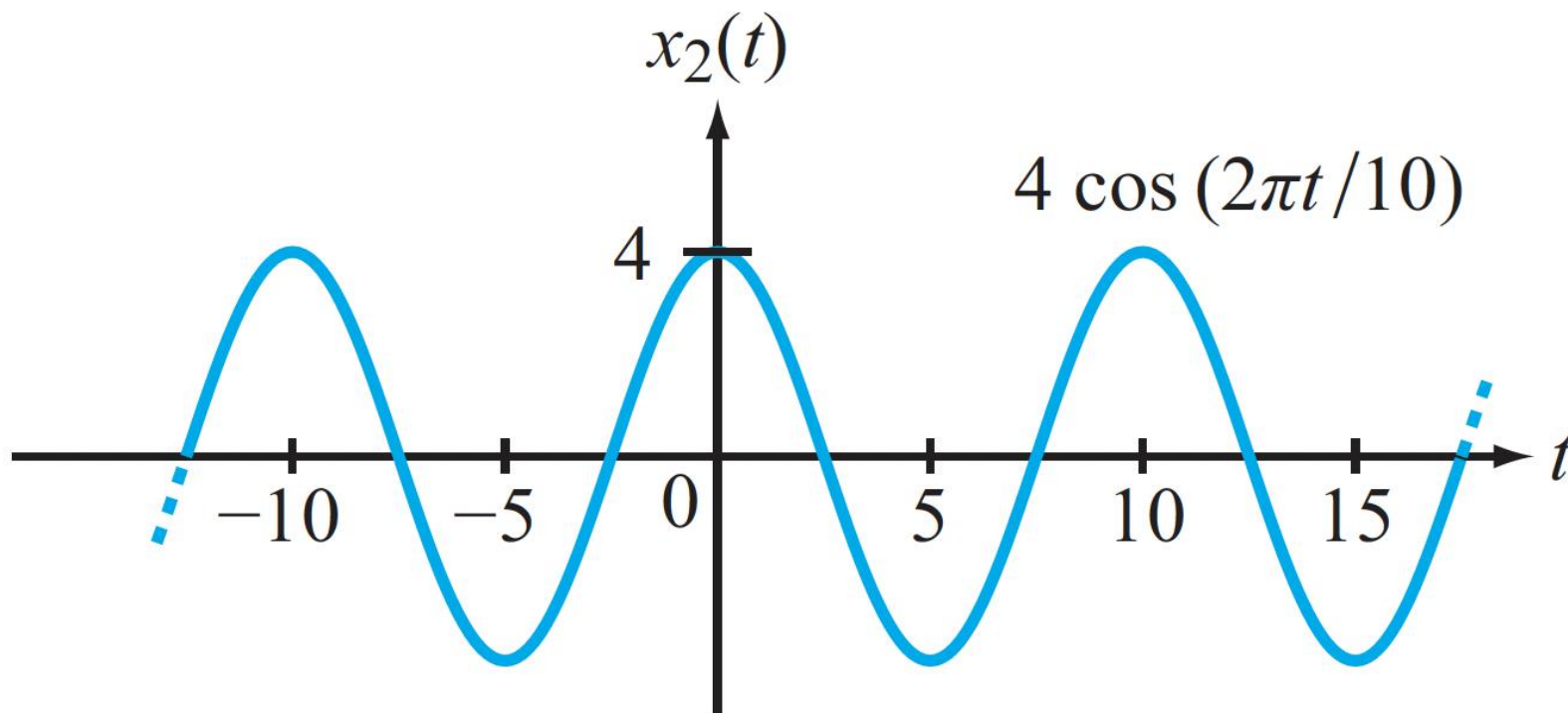
Recap: Basic Waveforms





Signal Power and Energy: Example 2

Calculate the power and energy for the following signal:



$$x_2(t) = 4 \cos\left(\frac{2\pi t}{10}\right)$$



Example 2 (contd)

$$x_2(t) = 4 \cos \left(\frac{2\pi t}{10} \right).$$

From the argument of $\cos(2\pi t/10)$, the period is 10 s. Hence, application of Eq. (1.36) leads to

$$\begin{aligned} P_{\text{av}2} &= \frac{1}{10} \int_{-5}^5 \left[4 \cos \left(\frac{2\pi t}{10} \right) \right]^2 dt \\ &= \frac{1}{10} \int_{-5}^5 16 \cos^2 \left(\frac{2\pi t}{10} \right) dt \\ &= 8. \end{aligned}$$

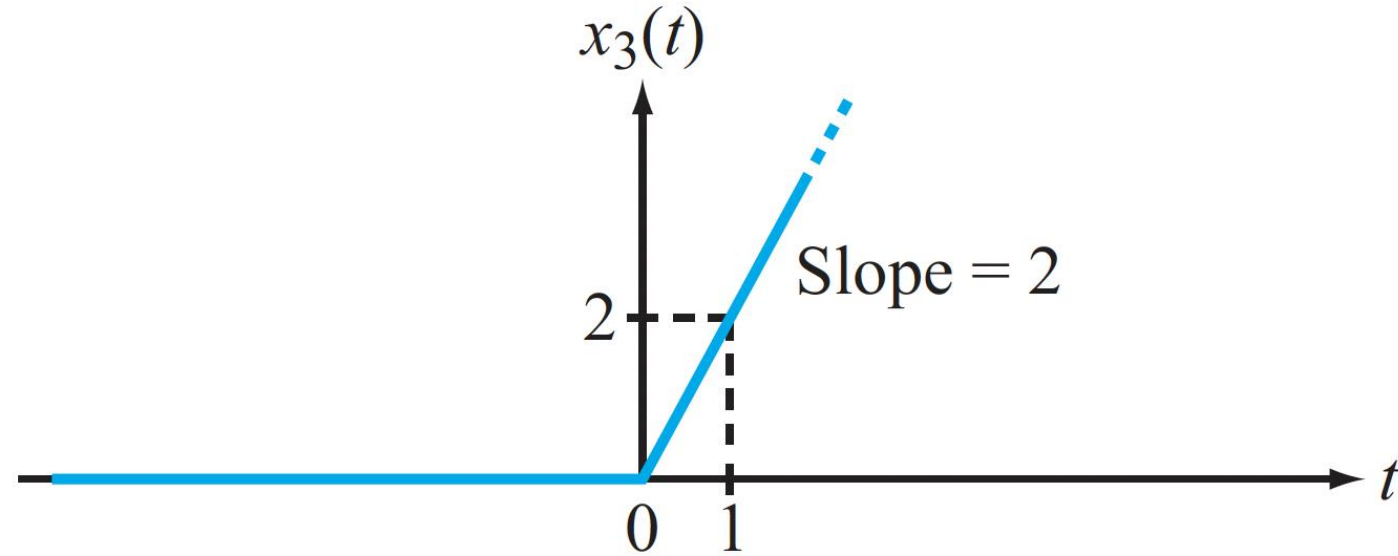
Since $P_{\text{av}2}$ is finite, total E_2 is infinite.



Example 3

The time-averaged power associated with $x_3(t)$ is

$$\begin{aligned} P_{\text{av}_3} &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^{T/2} 4t^2 dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \left[\frac{4t^3}{3} \Big|_0^{T/2} \right] \\ &= \lim_{T \rightarrow \infty} \left[\frac{1}{T} \times \frac{4T^3}{24} \right] \\ &= \lim_{T \rightarrow \infty} \left[\frac{T^2}{6} \right] \rightarrow \infty. \end{aligned}$$



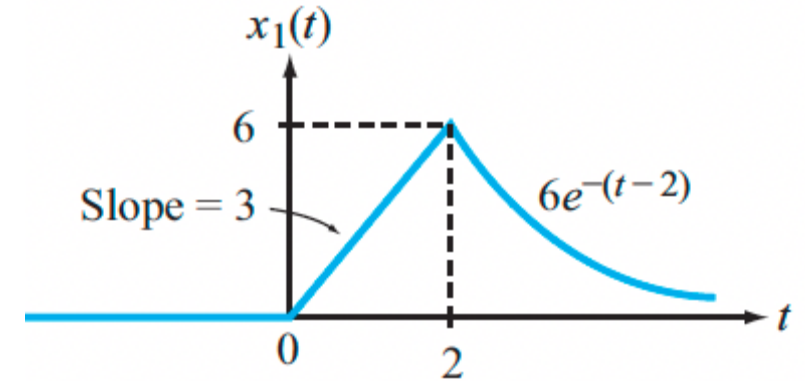
Moreover, $E_3 \rightarrow \infty$ as well.



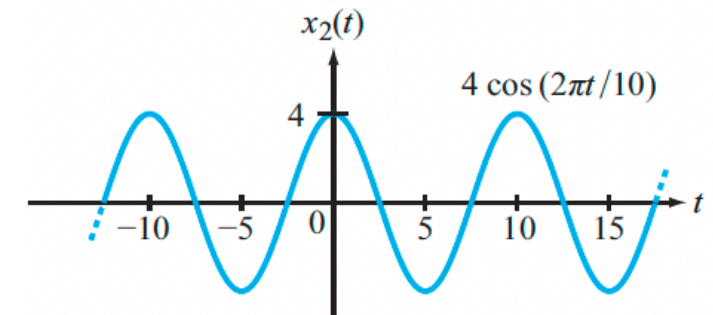
Three Types of Signals

Based on their power and energy profiles, there are 3 types of signals:

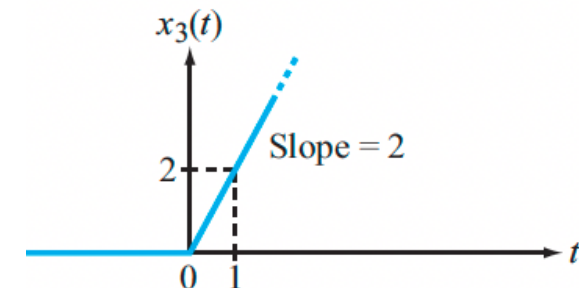
Energy signals: $P_{av} = 0$ and E is finite



Power signals: P_{av} is finite and E is infinite

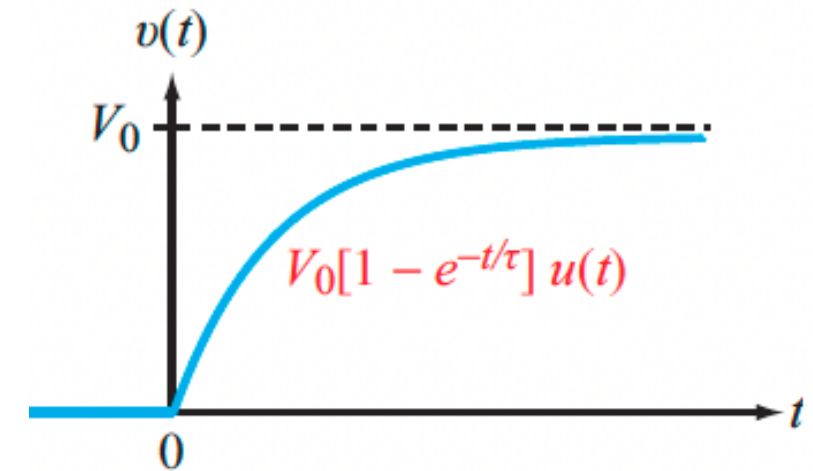
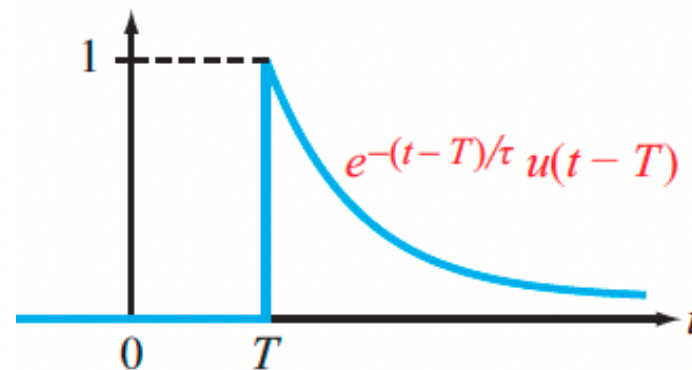
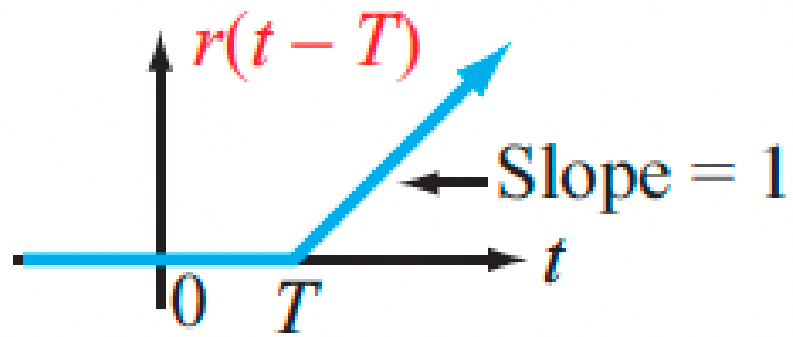
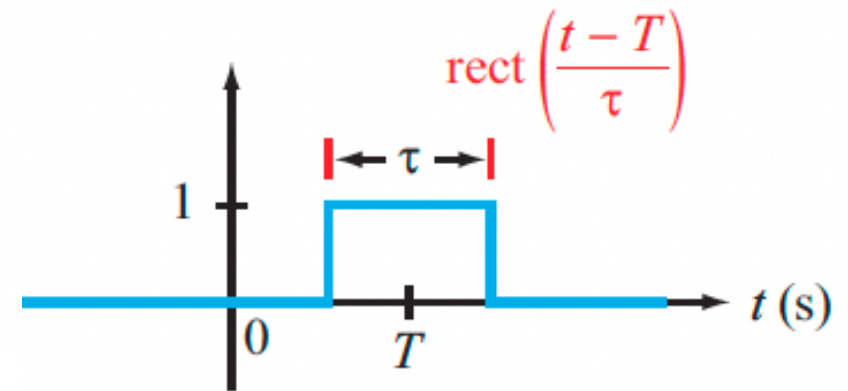
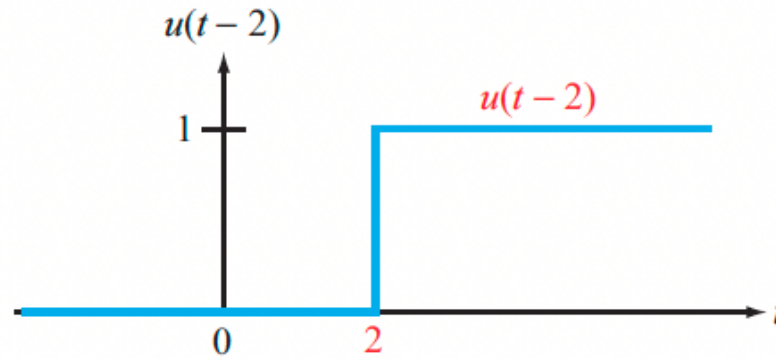
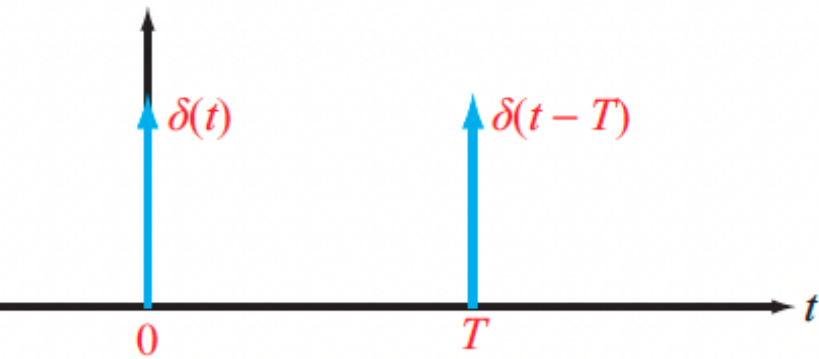


Non-physical signals: P_{av} and E are infinite





Summary of Basic Waveforms





Recap: Basic Formulas

$$\frac{du(t)}{dt} = \delta(t) \quad \frac{dr(t)}{dt} = u(t) \quad \delta(at) = \frac{1}{|a|} \delta(t)$$

$$\int_{-\infty}^t \delta(\tau) d\tau = u(t) \quad \int_{-\infty}^t u(\tau) d\tau = tu(t) = r(t)$$



Recap: Basic Formulas

$$x(t)\delta(t) = x(0)\delta(t)$$

$$\int_{-\infty}^{+\infty} \delta(t)s(t)dt = s(0) \int_{-\infty}^{+\infty} \delta(t)dt = s(0)$$

When the upper limit is the variable t , we get a function of t

$$\begin{aligned} \int_{-\infty}^t x(\tau)\delta(\tau)d\tau &= \int_{-\infty}^t x(0)\delta(\tau)d\tau \\ &= x(0) \int_{-\infty}^t \delta(\tau)d\tau = x(0)u(t) \end{aligned}$$



Please Sketch the Following

$$u(2t)$$

$$u(2t - 3)$$

Hint: Look for the value of t when $(2t - 3)$ crosses zero



Today: Getting to Know Systems



Well-designed systems have helpful properties:

- ✓ Stable operation (does not blow up)
- ✓ Predictable behaviors (no surprises)
- ✓ Ease of mathematical modeling (we can accurately calculate the outputs for the expected input signals)
- ✓ Ease of design to meet performance goals
- ✓ Ease of designing complex systems systematically, and on a large scale
- ✓ Ease of building, testing, and debugging systems
- ✓ Modularity (can we divide and conquer, mix and match parts etc.?)



How to Describe a System



A system can be specified (described/modeled) in a variety of ways

- Explicitly, as a function $y(t) = f(x(t))$
- Implicitly, as the solution of a differential equation, for example:

$$\frac{dy(t)}{dt} + y(t) = x(t)$$

- By specifying the system response to a standard “test signal” like $\delta(t)$ or $u(t)$

We will learn additional ways to describe the system, and ways to convert one type of representation into another



System Properties: Scaling

Many systems have the convenient “scaling property”:



Scaling the input results in an identical scaling of the output

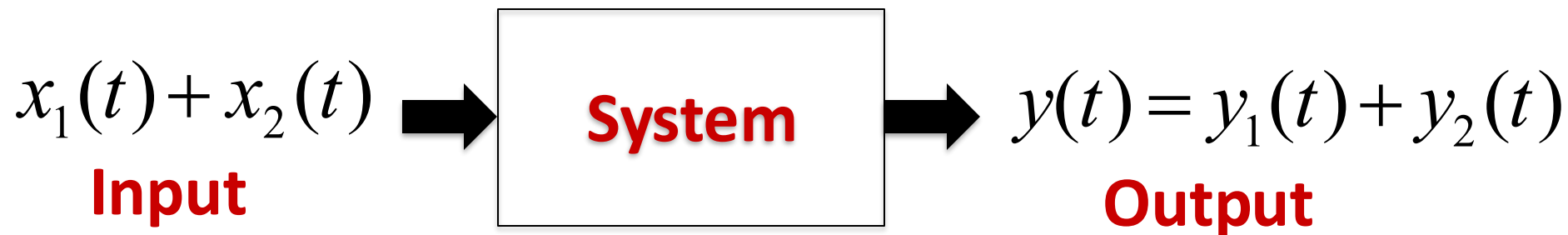
Example 1: Scaling Property: $y(t) = \frac{dx(t)}{dt} \quad \square \quad cy(t) = \frac{d}{dt}(cx(t))$

Example 2: No Scaling Property: $y(t) = x(t)^2$



System Property: Additivity

Many systems have another convenient property: Additivity



If we know the responses $y_1(t)$ and $y_2(t)$ to inputs $x_1(t)$ and $x_2(t)$, we can calculate the response to $[x_1(t) + x_2(t)]$.

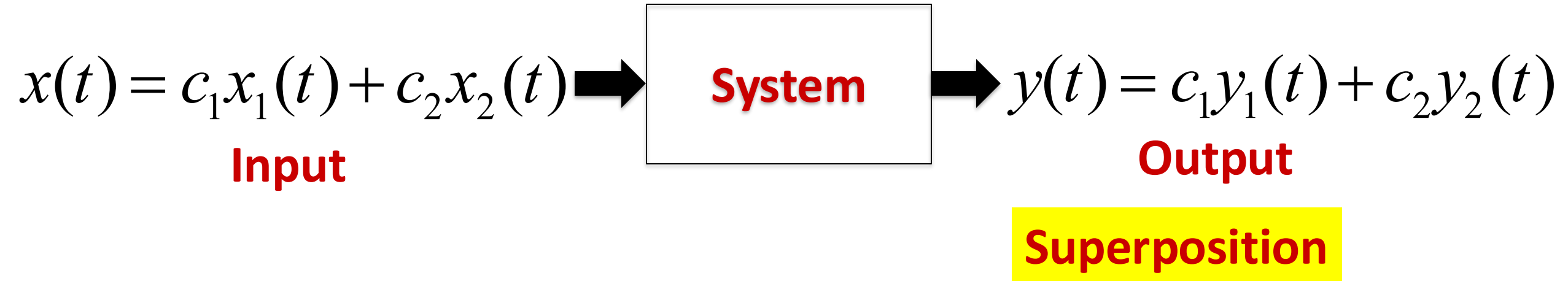
For example, this system is additive: $y(t) = \frac{dx(t)}{dt}$

$$y(t) = \frac{d[x_1(t) + x_2(t)]}{dt} = \frac{d[x_1(t)]}{dt} + \frac{d[x_2(t)]}{dt} = y_1(t) + y_2(t)$$



Linear Systems

Linear systems have both scaling and additivity properties:



Superposition is a big deal:

- Many complicated signals can be rewritten as a weighted sum of simpler signals whose responses can be calculated individually, and then superposed
- Many complicated systems are linear (e.g., RLC circuits)
- Many non-linear systems can be approximated by a linear model



Checking for Linearity

It is often easier to detect when a system is not linear

- To be linear, it must have **additivity** and **Scaling**
- If the system does not have any one of these properties, we can confidently say it is not linear

Example:

- Double the input, and see if the output doubles
- If it does not, then the system is clearly not linear!



Example

Check if the system $y(t) = 5x(t) + 3$ is linear

Step 1: Check for Scaling: Let $x_1(t) = cx(t)$
 $\Rightarrow y_1(t) = 5cx(t) + 3$

If the system was linear, the output should be $cy(t) = 5cx(t) + 3c$

This is clearly not the case, so this system is **NOT LINEAR**

There is no need to check for additivity



Example

Check if the system $\frac{dy(t)}{dt} + ty(t) = x(t)$ is linear

Step 1: Check for Scaling: Let $x_1(t) = cx(t)$

$$\Rightarrow \frac{dy_1(t)}{dt} + ty_1(t) = cx(t)$$

If the system was linear, the output $y_1(t)$ should be $cy(t)$

If we multiply the original system by c throughout, we get:

$$\frac{d[cy(t)]}{dt} + t[cy(t)] = cx(t) \text{ **OK**}$$

This system has the scaling property, so we can proceed to check if it also additive



Example

Check if the system $\frac{dy(t)}{dt} + ty(t) = x(t)$ is linear

Step 2: Check for Additivity:

$$\frac{dy_1(t)}{dt} + ty_1(t) = x_1(t)$$

$$\frac{dy_2(t)}{dt} + ty_2(t) = x_2(t)$$

$$\frac{d[y_1(t) + y_2(t)]}{dt} + t[y_1(t) + y_2(t)] = [x_1(t) + x_2(t)] \text{ OK}$$

Scaling and Additivity are BOTH True, so this system is LINEAR



Example

Check if the system $y(t) = \int_0^t x(\tau) d\tau$ is linear

Step 1: Check for Scaling: Let $x_1(t) = cx(t)$

$$\int_0^t x_1(\tau) d\tau = \int_0^t cx(\tau) d\tau = cy(t) \text{ OK}$$

Step 2: Check for Additivity:

$$\int_0^t [x_1(\tau) + x_2(\tau)] d\tau = \int_0^t x_1(\tau) d\tau + \int_0^t x_2(\tau) d\tau = y_1(t) + y_2(t) \text{ OK}$$

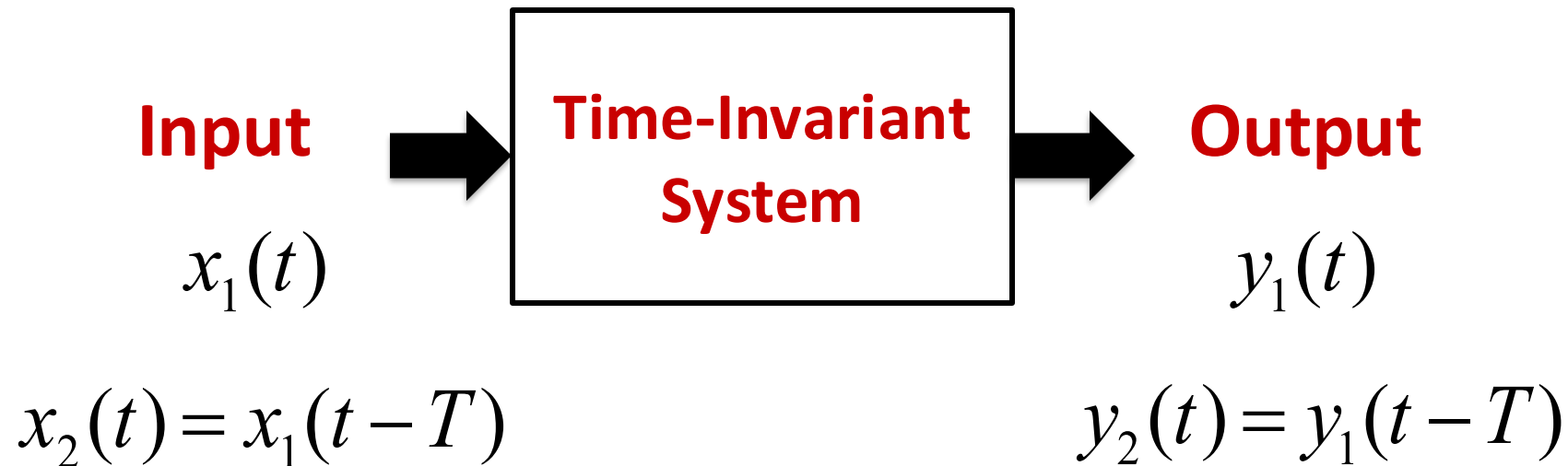
This system is LINEAR



Time-Invariant Systems

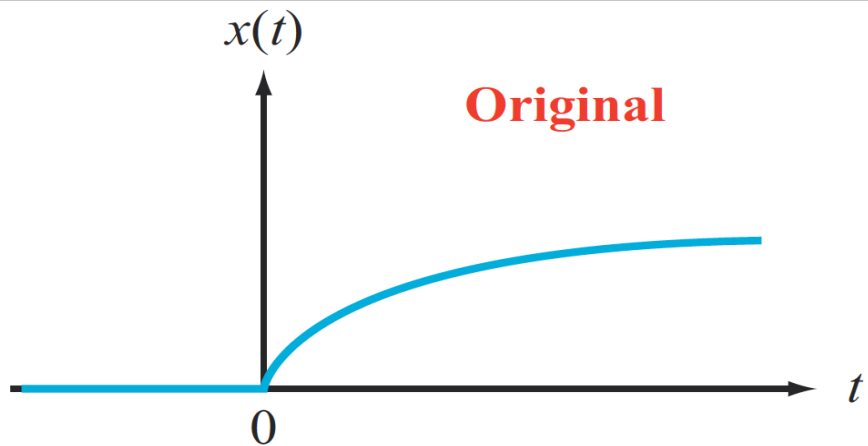
Many systems do not change over time

- Example: a fixed circuit built out of fixed-value resistors, capacitors, and inductors
- Delaying the input will delay the output by the same amount, with no other change

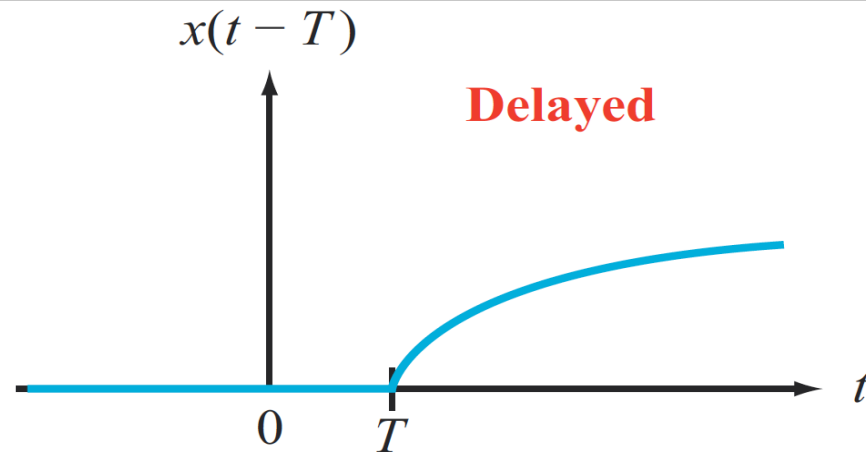




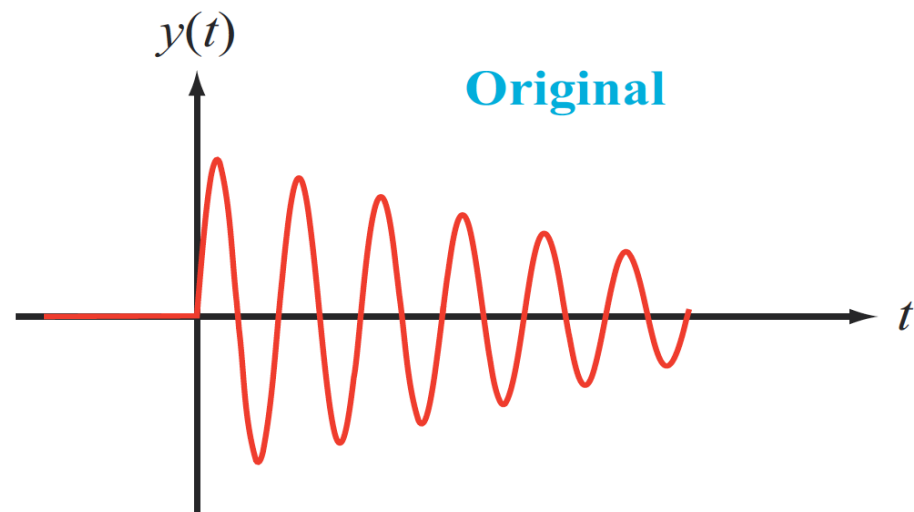
Example



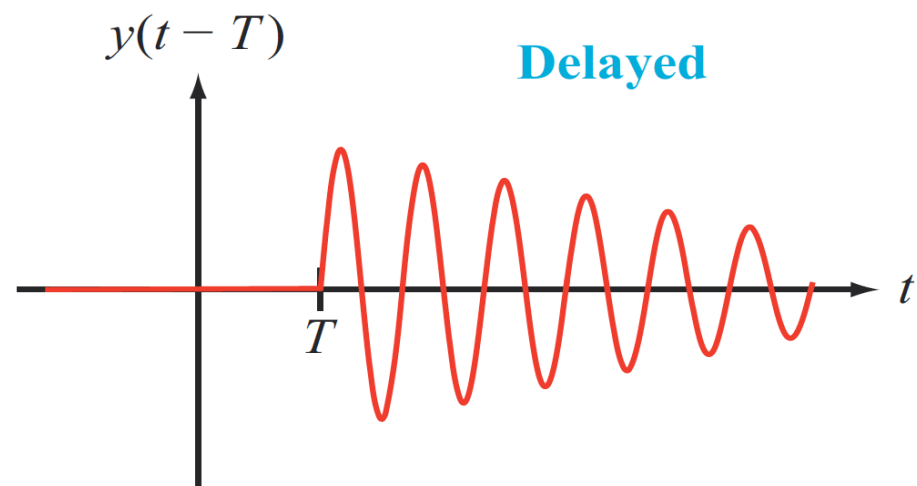
(a)



(c)



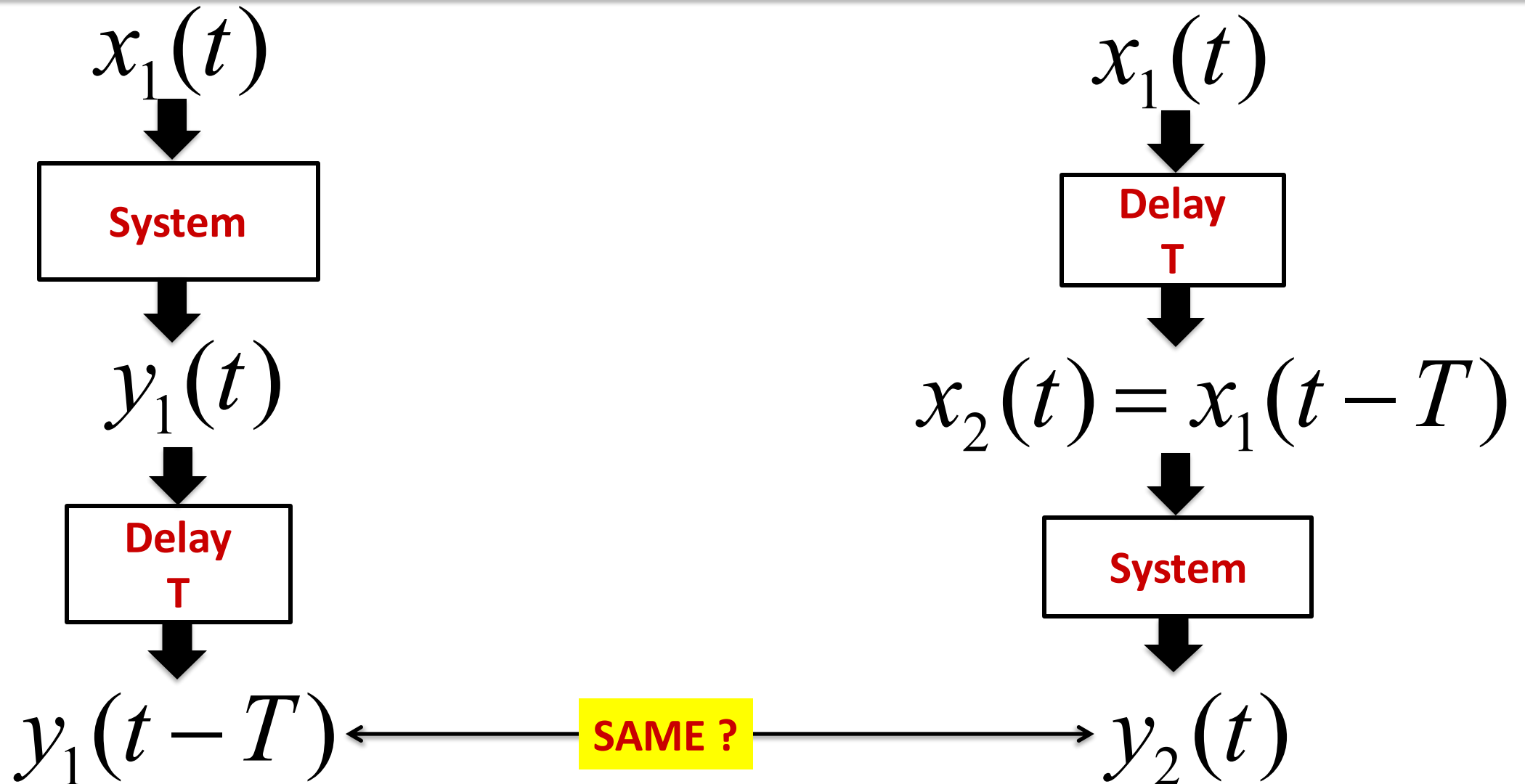
(b)



(d)



Checking for Time Invariance



IF THE SYSTEM AND DELAY ARE COMMUTATIVE THEN THE SYSTEM IS TIME INVARIANT



Example 1

$$y(t) = t x(t)$$

Multiply the input by t

$$y_1(t) = t x_1(t) \text{ System}$$

$$x_2(t) = x_1(t - T) \text{ Delay}$$

$$\square \quad y_1(t - T) = \underline{(t - T)x_1(t - T)}$$

Delay

$$\square \quad y_2(t) = tx_2(t) = \underline{tx_1(t - T)}$$

System

$$\text{Clearly, } y_2(t) \neq y_1(t - T)$$

Not time invariant!



Example 2

$$y(t) = x(t^2)$$

$$y_1(t) = x_1(t^2) \quad \text{System}$$

$$\square \quad y_1(t - T) = \underline{x_1((t - T)^2)} \quad \text{Delay}$$

$$x_2(t) = x_1(t - T) \quad \text{Delay}$$

$$\square \quad y_2(t) = x_2(t^2) = \underline{x_1(t^2 - T)} \quad \text{System}$$

Clearly, $y_2(t) \neq y_1(t - T)$ **Not time invariant!**



Example 3

$$y(t) = \sin(x(t))$$

$$y_1(t) = \sin(x_1(t)) \text{ System}$$

$$\square \quad y_1(t - T) = \underline{\sin(x_1(t - T))} \text{ Delay}$$

$$x_2(t) = x_1(t - T) \text{ Delay}$$

$$\square \quad y_2(t) = \sin(x_2(t)) = \underline{\sin(x_1(t - T))} \text{ System}$$

$$y_2(t) = y_1(t - T) \quad \text{Time invariant}$$



Example 4

$y(t) = x(-t)$ This system flips a signal in time, so t is replaced by $-t$

$y_1(t) = x_1(-t)$ System

□ $y_1(t - T) = x_1(-(t - T)) = \underline{x_1(-t + T)}$ Delay

$x_2(t) = x_1(t - T)$ Delay

□ $y_2(t) = x_2(-t) = \underline{x_1(-t - T)}$ System

$y_2(t) \neq y_1(t - T)$ NOT Time invariant



Example 5

$$y(t) = \int_0^t e^{-3\tau} x(\tau) d\tau$$

$$y_1(t) = \int_0^t e^{-3\tau} x_1(\tau) d\tau \quad \text{System}$$

$$\square \quad y_1(t-T) = \int_0^{t-T} e^{-3\tau} x_1(\tau) d\tau \quad \text{Delay}$$

$$x_2(t) = x_1(t-T) \quad \text{Delay}$$

$$\square \quad y_2(t) = \int_0^t e^{-3\tau} x_2(\tau) d\tau = \int_0^t e^{-3\tau} x_1(\tau-T) d\tau \quad \text{System}$$

$$y_2(t) \neq y_1(t-T) \quad \text{NOT Time invariant}$$



Another Thumb Rule

1) Shift only the signals by T :

$$x(t) \square x(t - T), y(t) \square y(t - T), \frac{d}{dt} y(t) \square \frac{d}{dt} y(t - T)$$

2) Replace all instances of t by $t - T$:

$$t \square t - T \quad \left(\text{don't change derivatives } \frac{d}{dt} \right)$$

3) If 1) = 2), this is time-invariant

Example: $y(t) = t x(t)$

Shift signals: $y(t - T) = t x(t - T)$

Replace $t \square t - T$: $y(t - T) = (t - T) x(t - T)$

Since: $t x(t - T) \square (t - T) x(t - T)$,

not time invariant

Example: $\frac{d}{dt} y(t) = x(t + 1)$

Shift signals: $\frac{d}{dt} y(t - T) = x(t - T + 1)$

Replace $t \square t - T$:

$$\frac{d}{dt} y(t - T) = x(t - T + 1)$$

Since: $x(t - T + 1) = x(t - T + 1)$

system is time invariant



Linear Time-Invariant Systems (LTI)

LTI Systems are both linear and time-invariant

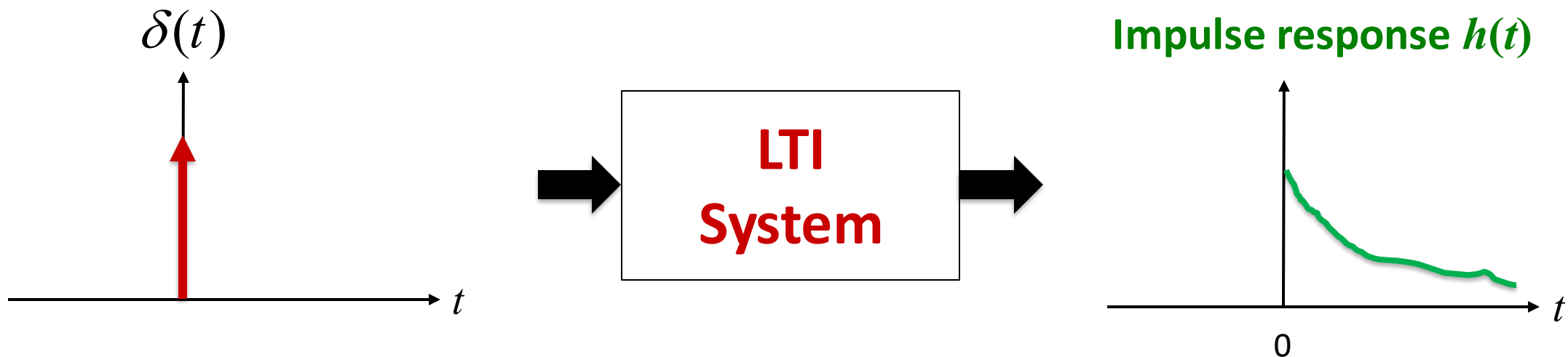
- Superposition (scaling & additivity), and
- Time invariance

This combination of properties allows simple analysis of remarkably complicated systems

- Non-LTI systems are much harder to analyze
- For this reason, we often approximate non-LTI systems by LTI models so we can analyze them



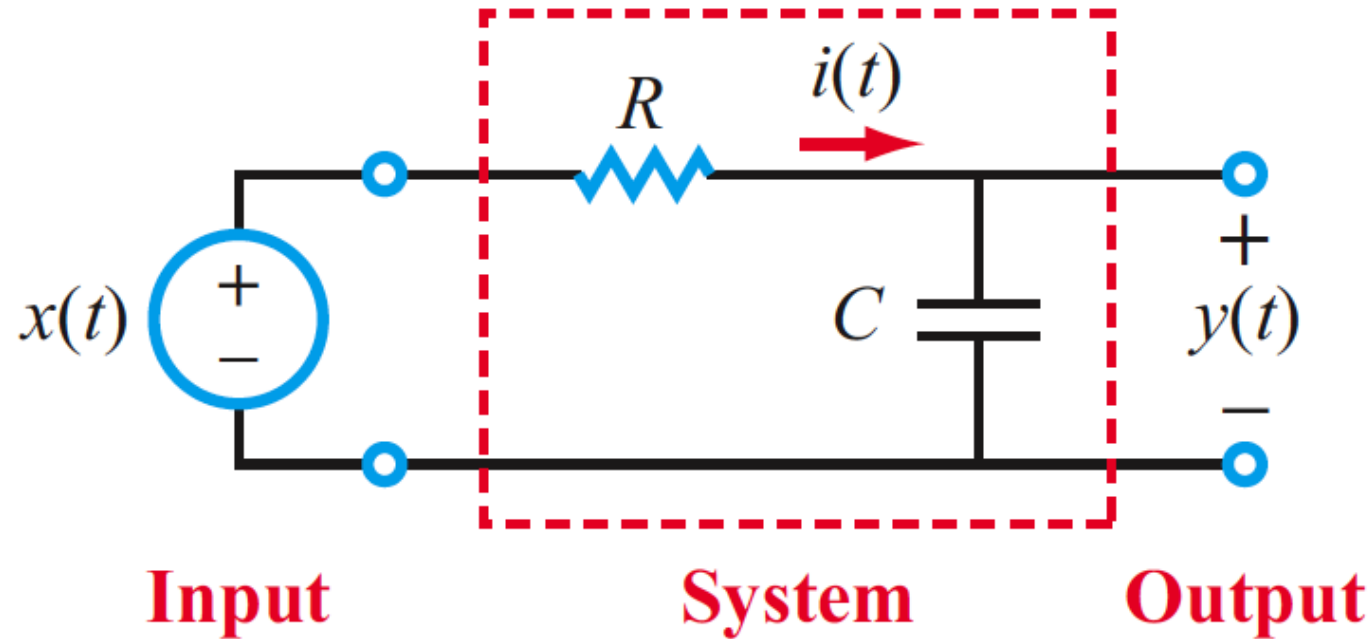
The Impulse Response



We can think of the impulse $x(t) = \delta(t)$ as a “**test signal**” that elicits a response $y(t) = h(t)$ that is informative of the system, just like kicking a car’s tires informs us about their inflation level



Example: An RC Circuit as a System



How can we characterize this system?



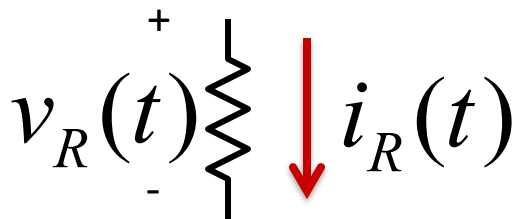
Recap of Circuit Elements



V

Voltage Source

(zero resistance, infinite current capability)



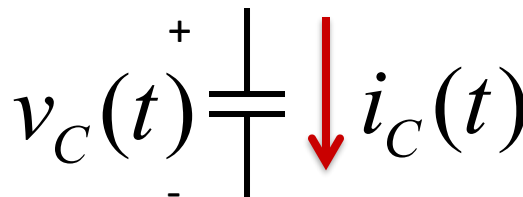
$$v_R(t) = i_R(t) R$$



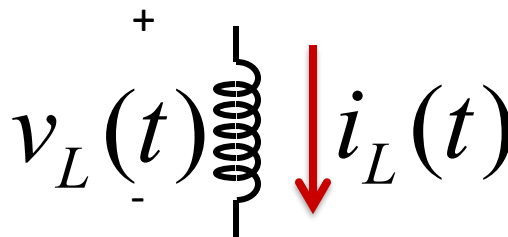
I

Current Source

(infinite resistance, infinite voltage capability)



$$i_C(t) = C \frac{dv_C(t)}{dt}$$



$$v_L(t) = L \frac{di_L(t)}{dt}$$



Example: Impulse Response of an RC Circuit

Kirchoff's Voltage Equation (KVL)

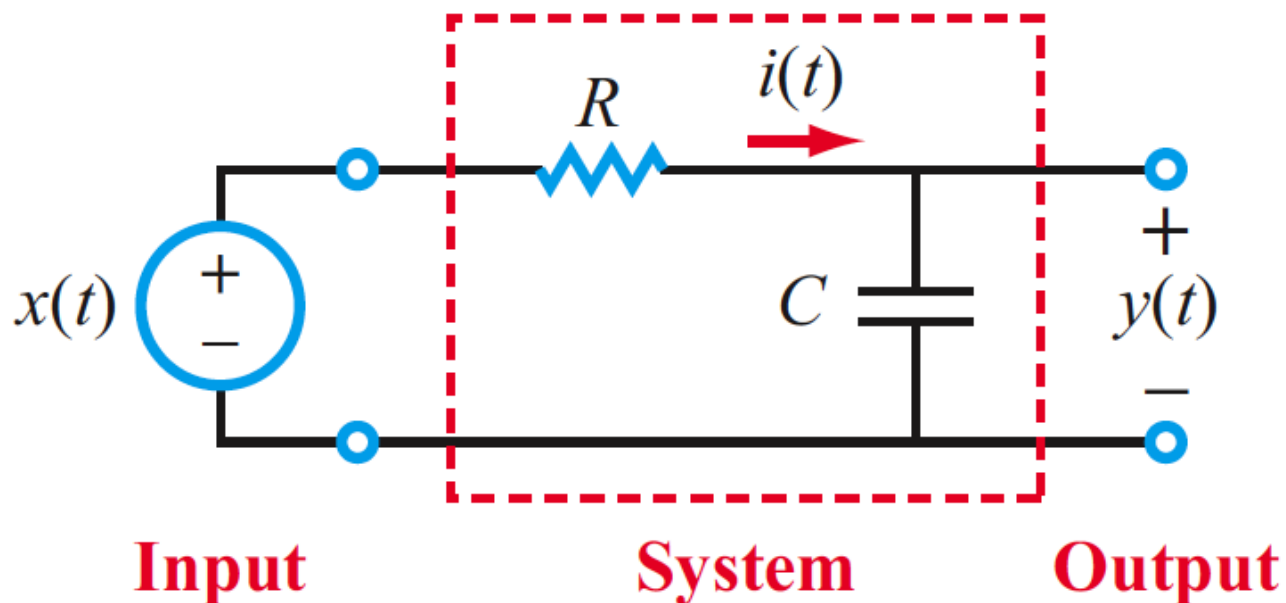
$$R i(t) + y(t) = x(t)$$

Capacitor Equation $i(t) = C \frac{dy}{dt}$

$$\Rightarrow RC \frac{dy}{dt} + y(t) = x(t)$$

Rewrite in standard form

$$\frac{dy}{dt} + \frac{1}{RC} y(t) = \frac{1}{RC} x(t)$$



Set $x(t) = \delta(t)$ (Impulse Input)

$$\frac{dy}{dt} + \frac{1}{RC} y(t) = \frac{1}{RC} \delta(t)$$

Need to solve this differential equation



Recap of Differential Equations

To solve the linear differential equation: $\frac{dy}{dx} + P(x)y = Q(x)$

Multiply both sides by the **integrating factor** $e^{\int P(x)dx}$ and integrate both sides

$$\underbrace{e^{\int P(x)dx} \frac{dy}{dx} + P(x)e^{\int P(x)dx} y}_{\text{derivative of a product}} = Q(x)e^{\int P(x)dx}$$

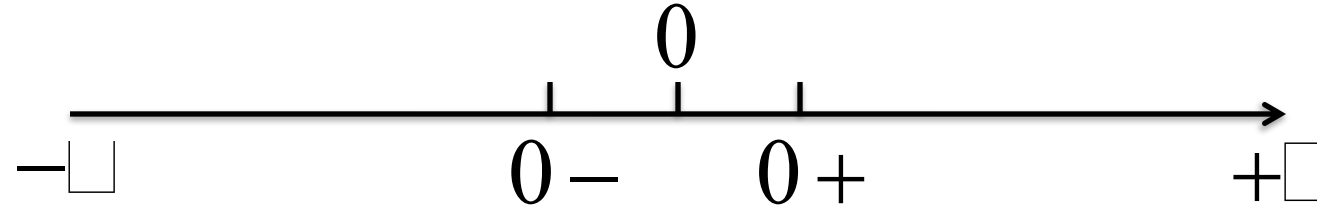
Now, recognize the left-hand side as the derivative of a product of two functions

$$\underbrace{e^{\int P(x)dx} \frac{dy}{dx} + P(x)e^{\int P(x)dx} y}_{\text{derivative of a product}} = \frac{d}{dx} \left\{ y e^{\int P(x)dx} \right\}$$

Integrate both sides to derive the answer



Recap of Calculus Notation



0 is the origin

“0 -” is infinitesimally to the left of the origin

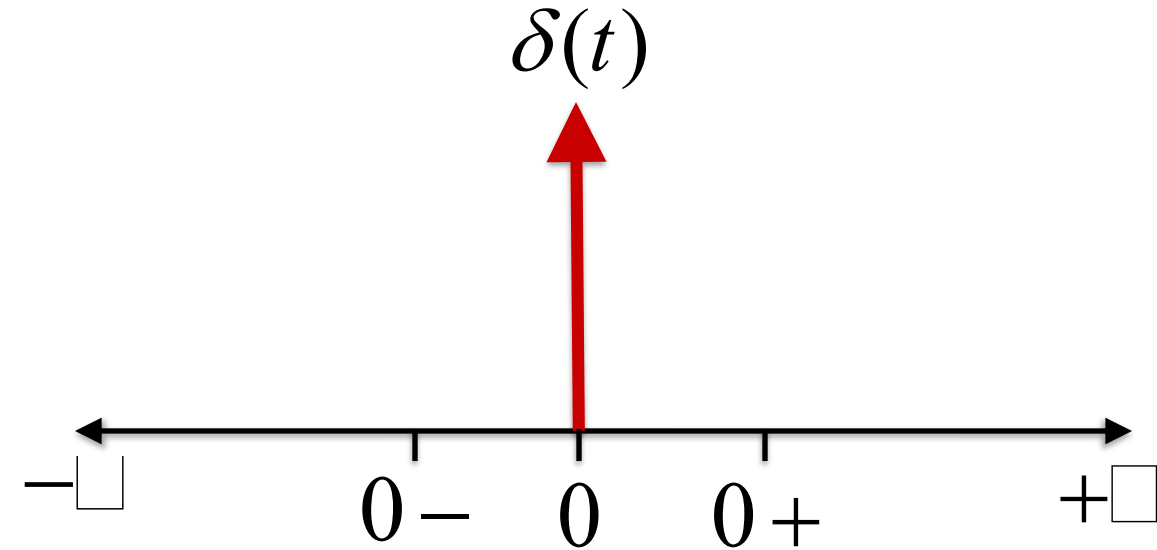
$$\lim_{\varepsilon \rightarrow 0} \{-\varepsilon\}, \varepsilon > 0$$

“0 +” is infinitesimally to the right of the origin

$$\lim_{\varepsilon \rightarrow 0} \{\varepsilon\}, \varepsilon > 0$$



Integrating an Impulse



$$\int_{0-}^{0+} \delta(t) dt = \int_{0-}^1 \delta(t) dt = \int_{0-}^{\infty} \delta(t) dt = 1$$
$$\int_{0-}^{\infty} \delta(t) dt = u(t)$$

WHILE SOLVING THE DIFFERENTIAL EQUATION, WE NEED TO START OUR INTEGRATION FROM “0–” TO INCLUDE THE IMPULSE INPUT AT 0 IN OUR CALCULATIONS. OTHERWISE, WE WILL MISS IT



RC Example Continued

$$\frac{dy}{dt} + \frac{1}{RC}y(t) = \frac{1}{RC}\delta(t)$$

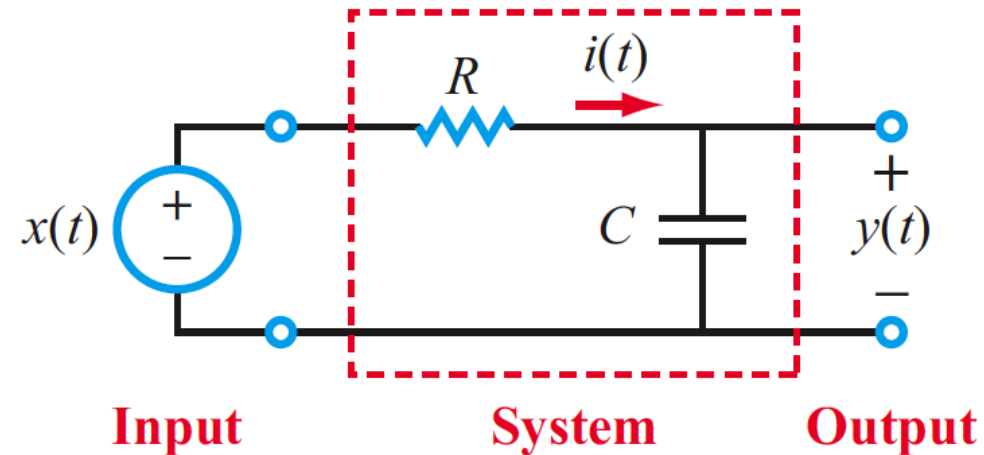
INTEGRATING FACTOR: $e^{\int \frac{1}{RC} dt} = e^{t/RC}$

$$\frac{dy}{dt} e^{t/RC} + \frac{1}{RC} e^{t/RC} y(t) = \frac{1}{RC} \delta(t) e^{t/RC}$$

$$\frac{dy}{dt} \{y(t) e^{t/RC}\} e^{t/RC} = \frac{1}{RC} \delta(t) e^{t/RC} = \frac{1}{RC} \delta(t) e^{0/RC} = \frac{1}{RC} \delta(t)$$

$$y(t) e^{t/RC} \Big|_{0^-}^t = \int_{0^-}^t \frac{1}{RC} \delta(t) dt = \frac{1}{RC} u(t) \quad \text{Integrate both sides from } 0^- \text{ to } t$$

$$\Rightarrow y(t) e^{\frac{t}{RC}} - y(0^-) = \frac{1}{RC} u(t) \quad \text{Valid only for } t > 0^-$$





RC Example Continued

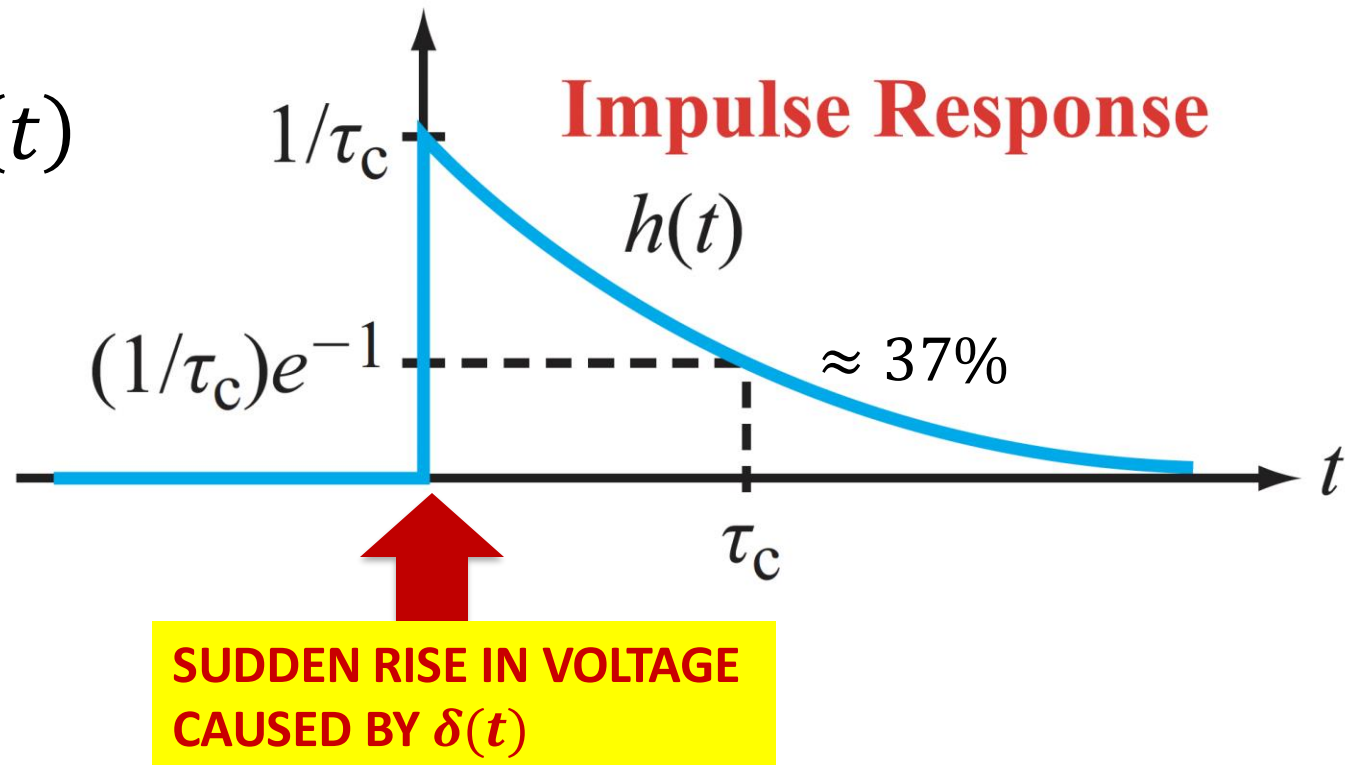
Rewrite for all t

$$y(t)e^{\frac{t}{RC}} = [y(0-) + \frac{1}{RC}]u(t)$$

When initial output $y(0-) = 0$

$$\begin{aligned} y(t) &= \frac{1}{RC} e^{-t/RC} u(t) \\ &= h(t) \end{aligned}$$

IMPULSE RESPONSE
(STANDARD NOTATION)



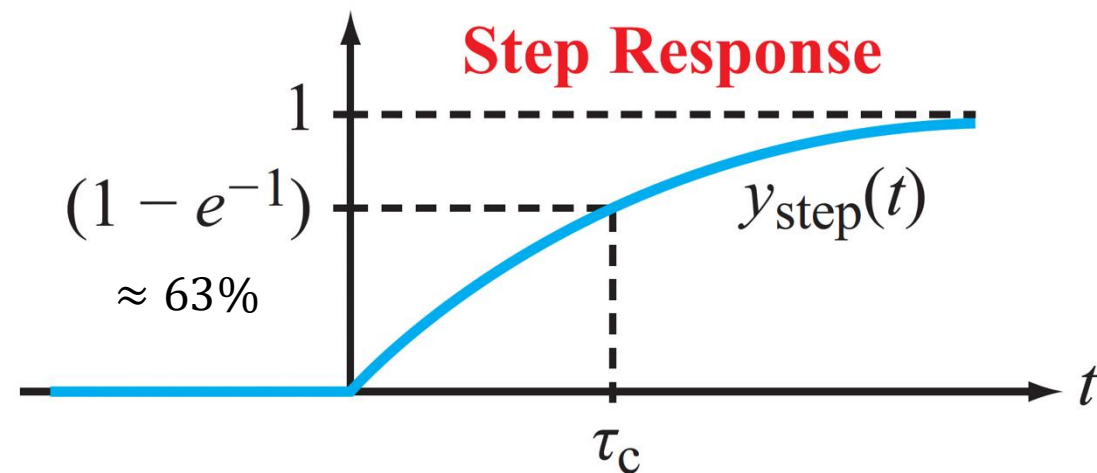
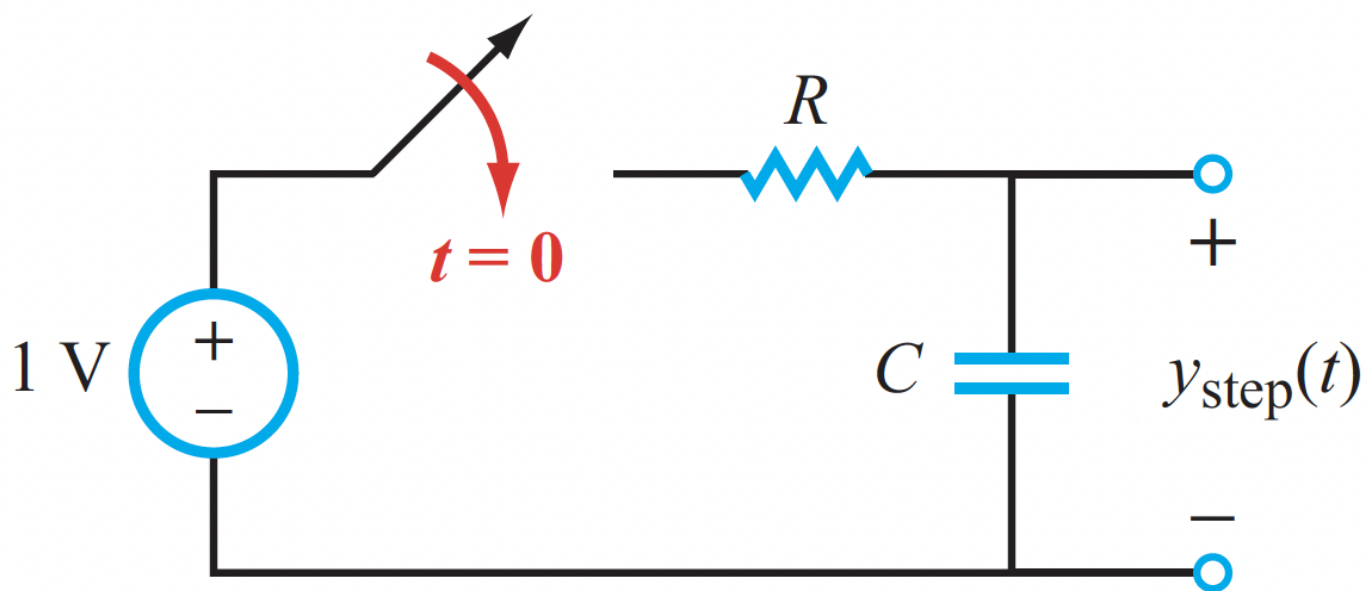
time constant $\tau_c = RC$

$$h(t) = 0 \text{ for } t < 0$$



RC Example Continued

Let's calculate the unit step response instead, so $x(t) = u(t)$



time constant $\tau_c = RC$



RC Example Continued

Let's calculate the unit step response instead, so $x(t) = u(t)$

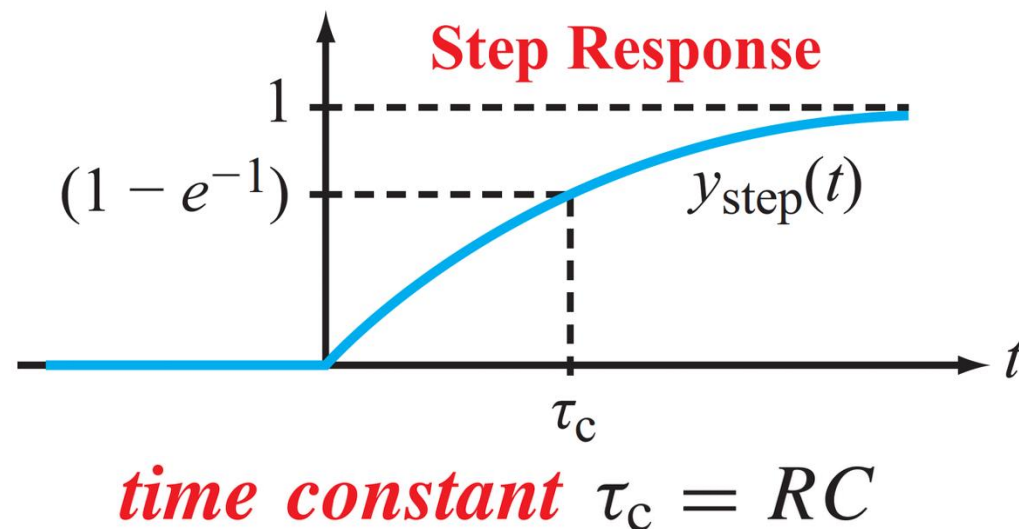
$$y_{Step}(t)e^{t/RC} = \int_{0-}^t \frac{1}{RC} e^{\tau/RC} \underline{u(\tau)} d\tau$$

REPLACE IMPULSE
WITH STEP

$$y_{Step}(t)e^{t/RC} = \frac{1}{RC} \int_{0-}^t e^{\tau/RC} u(\tau) d\tau \quad \text{FOR } t > 0$$

$$= \frac{1}{RC} \left\{ RC \left[e^{\tau/RC} \right]_{0-}^t \right\} u(t) = \{e^{t/RC} - 1\} u(t) \quad u(t) \text{ ENFORCES } t > 0$$

$$\int y_{Step}(t) = e^{-t/RC} \{e^{t/RC} - 1\} u(t) = \{1 - e^{-t/RC}\} u(t) \quad \text{STEP RESPONSE WHEN } y(0-) = 0$$





How do They Relate?

We have seen that:

$$\begin{aligned}h(t) &= \frac{1}{RC} e^{-t/RC} u(t) & h(t) &= \frac{d}{dt} \{y_{STEP}(t)\} = \frac{d}{dt} \{(1 - e^{-t/RC})u(t)\} \\y_{Step}(t) &= (1 - e^{-t/RC})u(t) & &= \frac{d}{dt} \{u(t) - e^{-t/RC}u(t)\} = \delta(t) - \frac{d}{dt} \{e^{-t/RC}u(t)\} \\& & &= \delta(t) - \{e^{-t/RC} \left[-\frac{1}{RC}u(t) + e^{-t/RC} \delta(t) \right]\} \\& & &= \cancel{\delta(t)} + \frac{1}{RC} e^{-t/RC} u(t) - e^{-0/RC} \cancel{\delta(t)} = \frac{1}{RC} e^{-t/RC} u(t)\end{aligned}$$

THIS IS TRUE GENERALLY FOR LTI SYSTEMS, $\mathbf{h(t) = \frac{dy_{STEP}(t)}{dt}}$



RC Time Constant

The RC product term appeared throughout our example

$$\tau_C = RC$$

It has a name: “**The RC Time Constant**”, and it is measured in seconds

$$h(t) = \frac{1}{RC} e^{-t/RC} u(t) = \frac{1}{\tau_C} e^{-t/\tau_C} u(t)$$

$$y_{Step}(t) = (1 - e^{-t/\tau_C}) u(t)$$



Practical Perspective

In the lab, we can generate pulses using a “function generator”

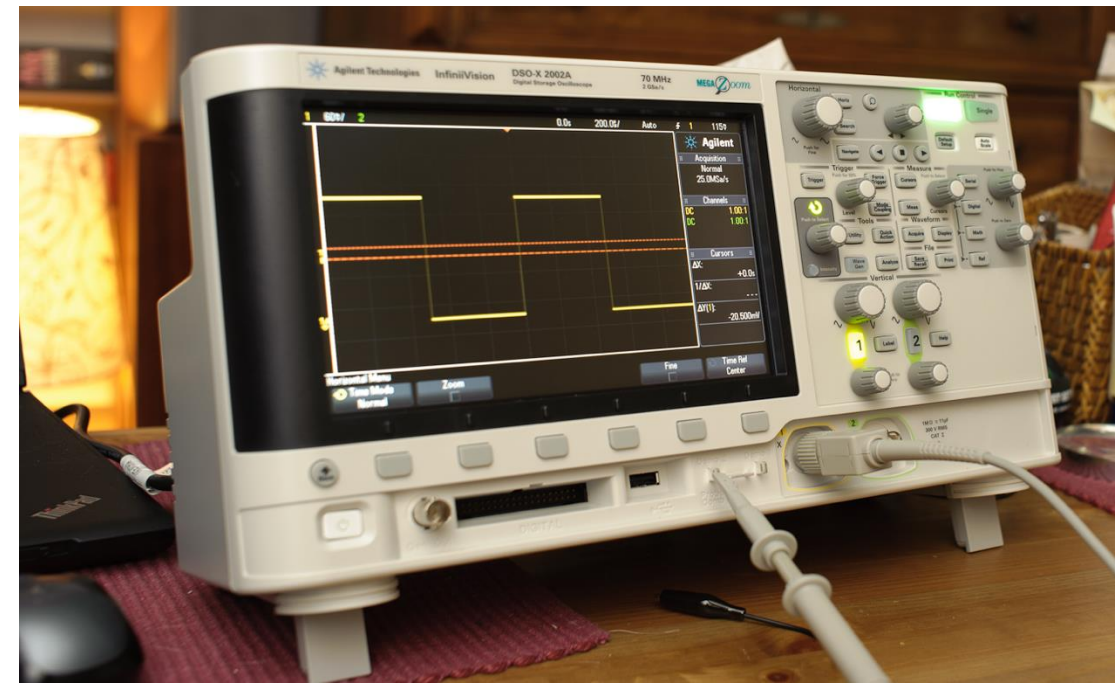
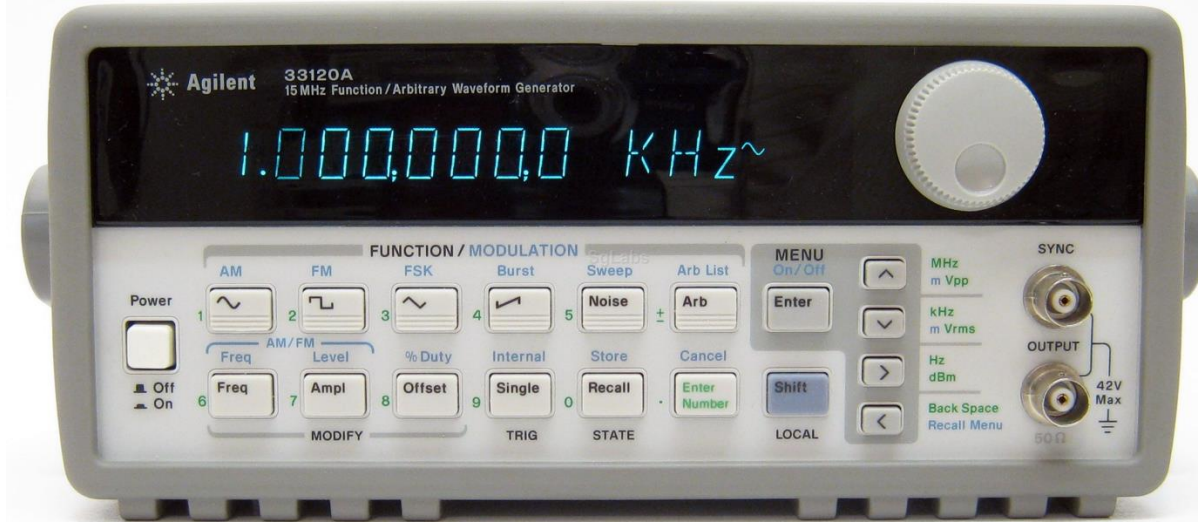
- The time period, pulse width, and height can be adjusted independently

We can view signals on an “oscilloscope”

- Observing the pulse response for a circuit is very informative

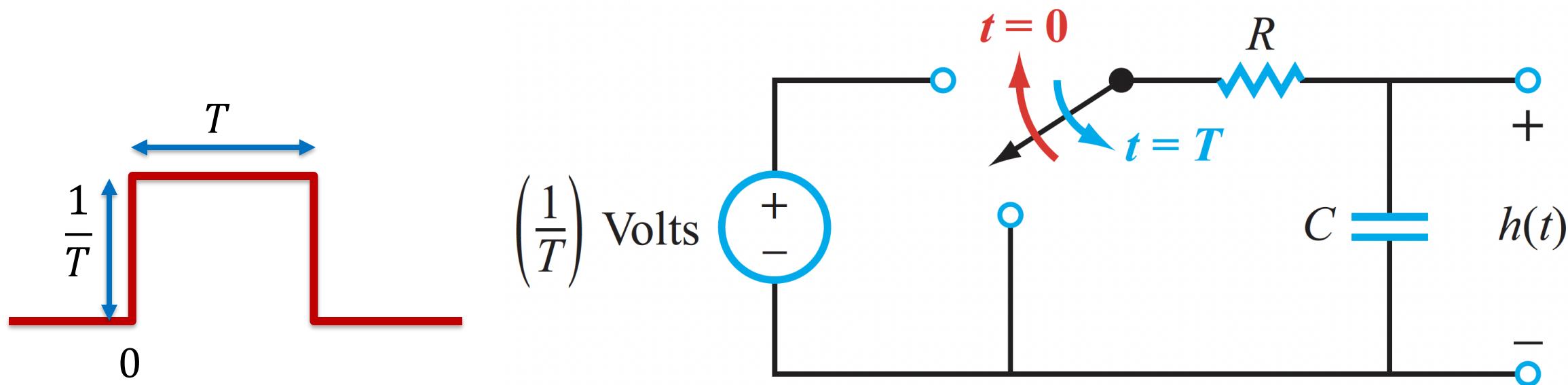
A tall and narrow pulse can be a decent approximation to an impulse

- *The pulse width should be much narrower (1/10) than the RC time constant!*





Practical Approximation of an Impulse



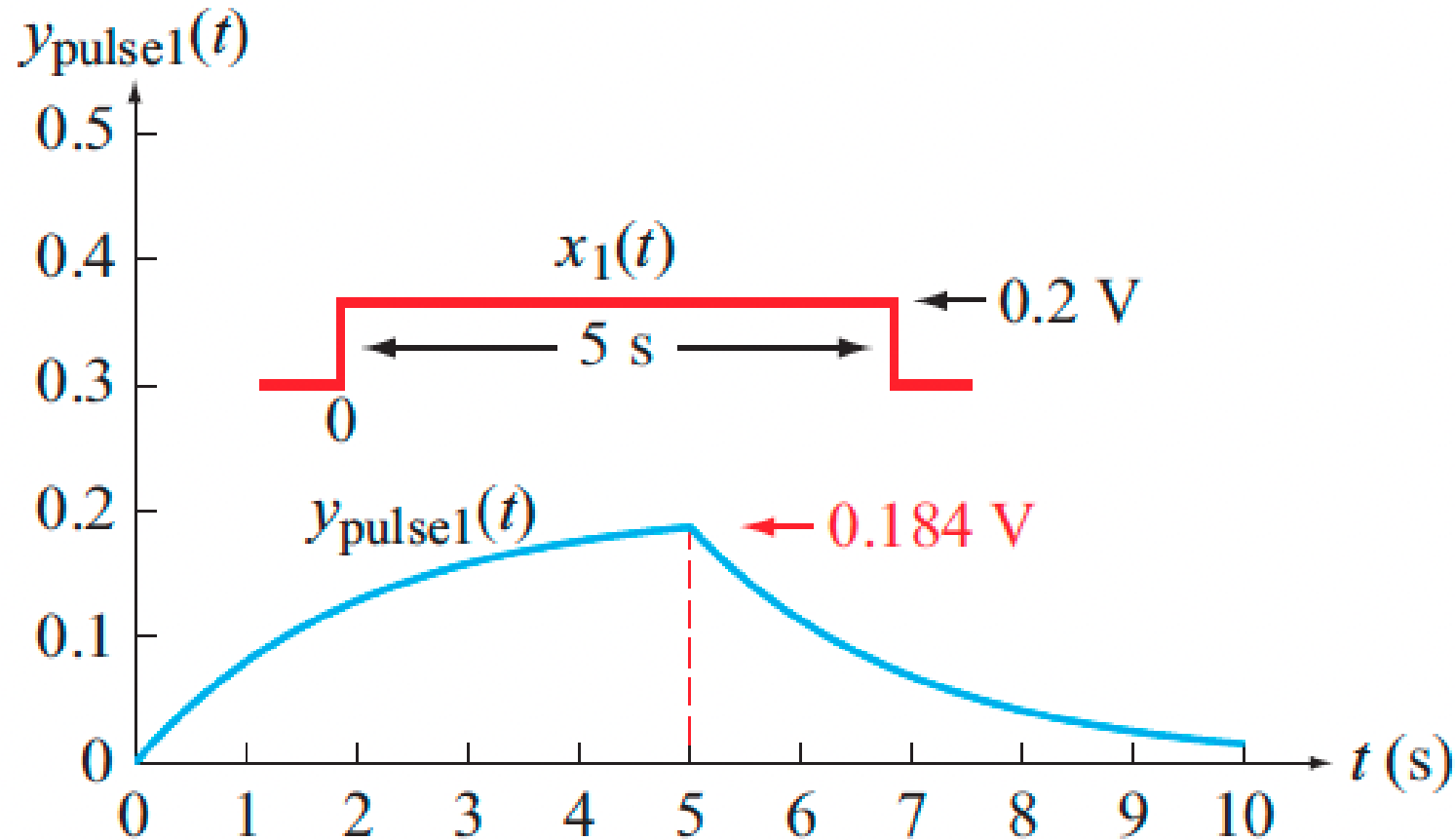
$$\delta(t) \square x_{PULSE}(t) = \frac{1}{T} [u(t) - u(t - T)]$$

$$y_{PULSE}(t) = \frac{1}{T} [y_{STEP}(t) - y_{STEP}(t - T)] \quad \text{LINEARITY}$$

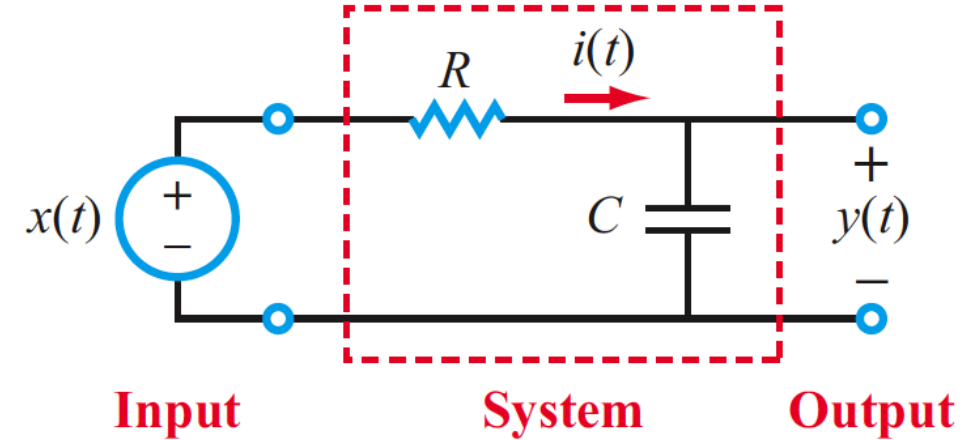
By reducing T , we can make a better approximation to $\delta(t)$



Pulse Input, $T = 5$ sec, $RC = 2$ sec



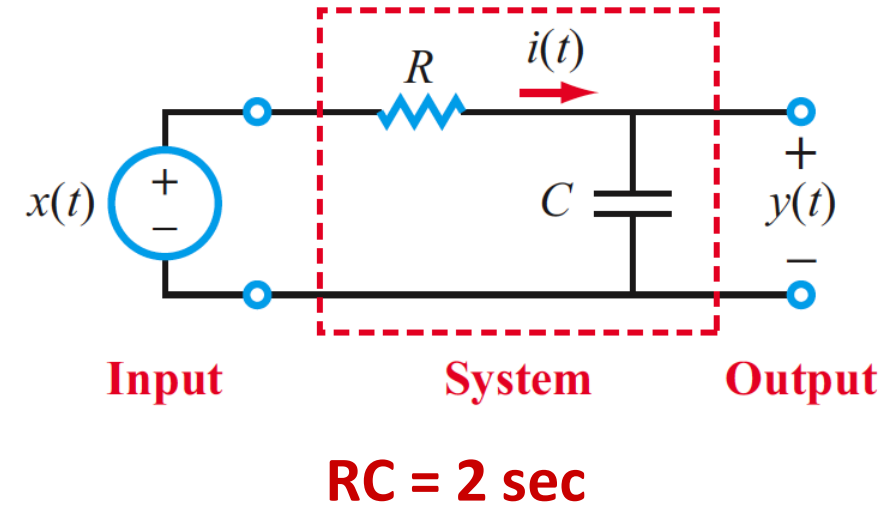
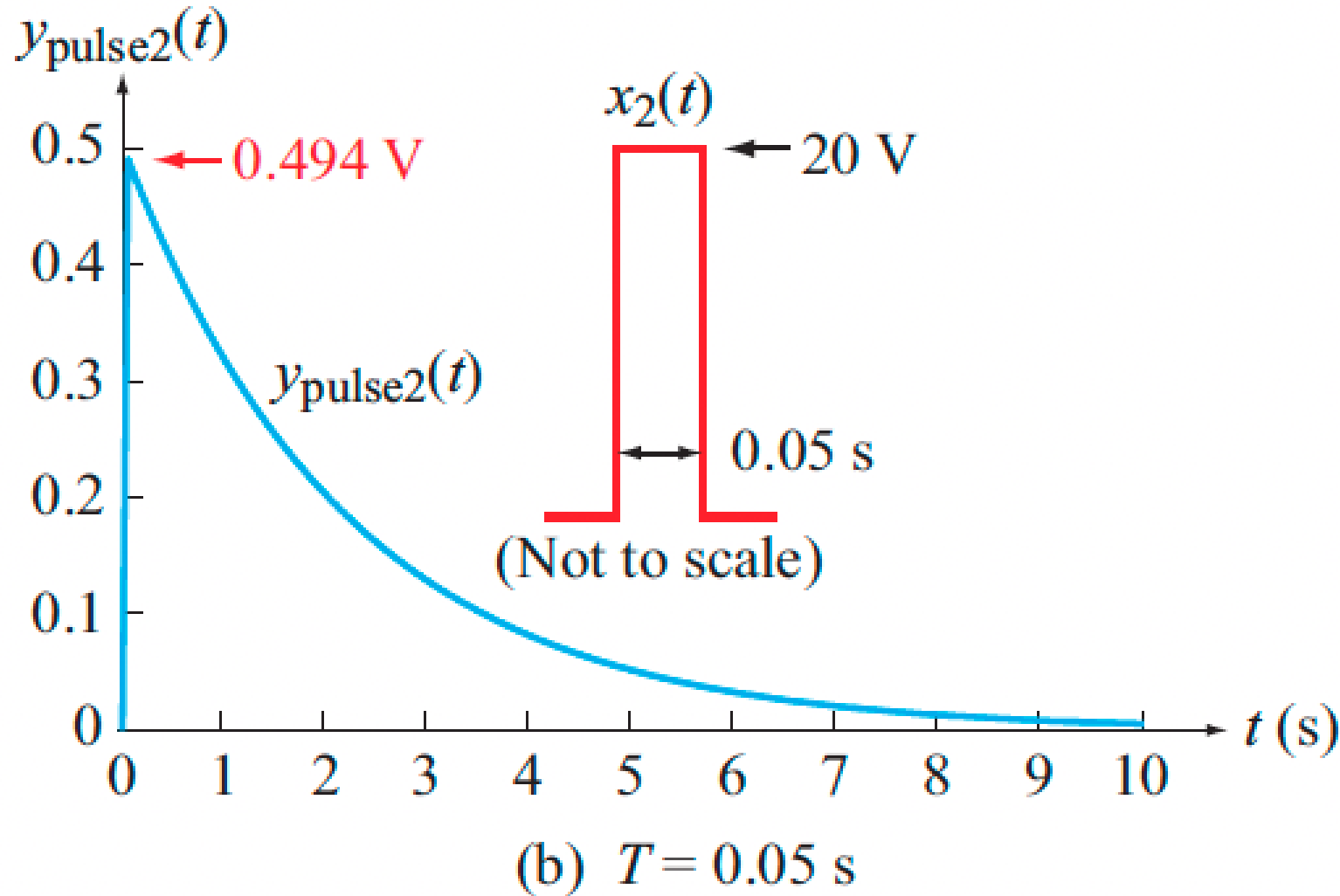
(a) $T = 5$ s



$$RC = 2 \text{ sec}$$



Narrower Pulse, $T = 0.05$ sec , $RC = 2$ sec

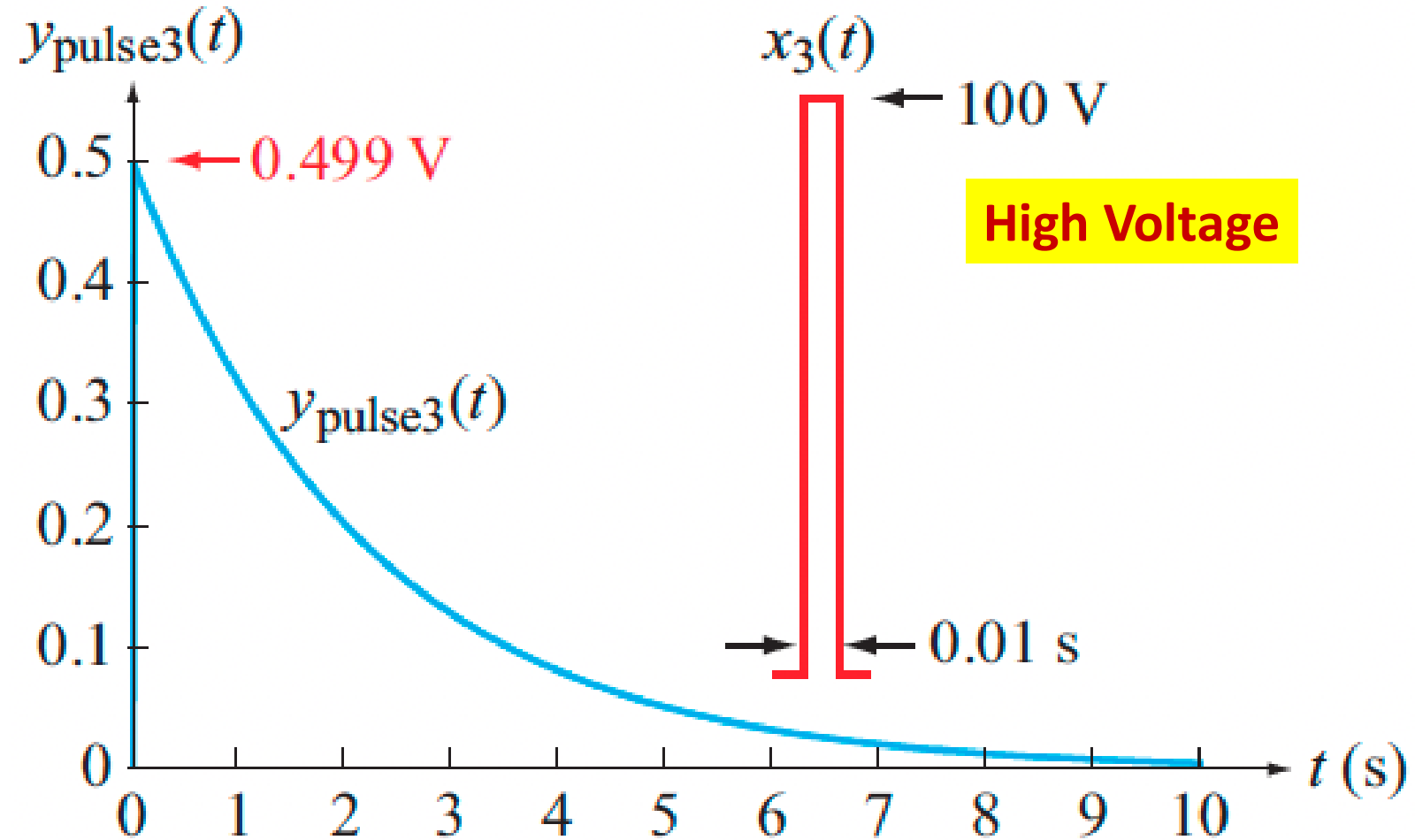


$$y_{pulse2}(t) \approx \frac{1}{RC} e^{-\frac{t}{RC}} u(t)$$

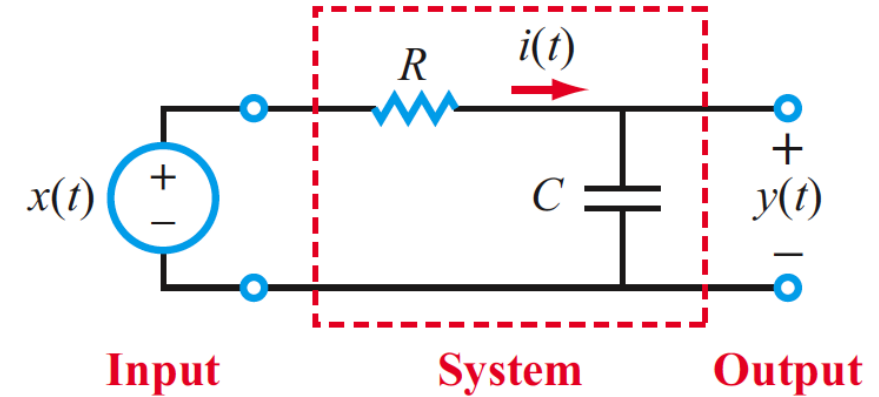
**Decent
approximation!**



Even Narrower, $T = 0.01$ sec , $RC = 2$ sec



(c) $T = 0.01$ s



$RC = 2$ sec

$$y_{pulse3}(t) \approx \frac{1}{RC} e^{-\frac{t}{RC}} u(t)$$

**Awesome
Approximation!!**



Practical Perspective

Generating a high-voltage, narrow-width pulse can be problematic, so a better strategy is desirable



- We measure the unit step response ($y_{STEP}(t)$) of the given system instead
- This is safer since it does not require a high voltage
- It is also simpler because it does not require rapidly changing waveforms
- We differentiate the measured $y_{STEP}(t)$ to figure out $h(t)$

$$h(t) = \frac{d}{dt} \{y_{STEP}(t)\}$$



Going Further!

Since a ramp is the integral of the unit step, we can also derive the response to a ramp from $y_{step}(t)$:

$$r(t) = \int_{-\infty}^t u(\tau) d\tau$$

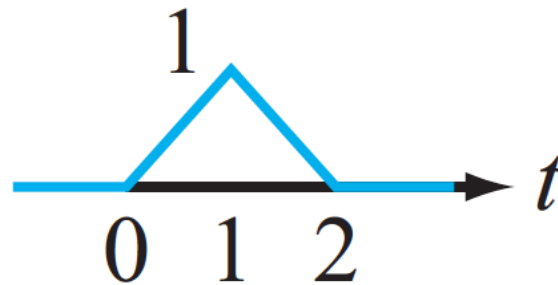
$$y_{ramp}(t) = \int_{-\infty}^t y_{step}(\tau) d\tau$$

Question: How can you compute the impulse response from the ramp response?



Try these Problems Now

1. Check the system $y(t) = x(t)u(t)$ for Linearity and Time Invariance
2. What is the impulse response $h(t)$ of a system whose step response $y_{STEP}(t)$ is the following:





Summary



LTI Systems are both linear and time-invariant

- Superposition (scaling & additivity), and
- Time invariance
- Important, Pervasive

We learned to analyze an RC circuit to calculate its impulse response $h(t)$ & step response $y_{STEP}(t)$

- Happily, this system is LTI 😊



For the Next Class

Work through the textbook examples in Sections 2.1 – 2.2

Study Section 2.3 on Zybooks, and do the assigned activities

(Remember that 10% of your course grade is for completing the Zybook activities in advance of each lecture)

How to Study:

- 1. Read the assigned sections to understand the basic concepts, and aim to memorize the concepts**
- 2. Attempt the in-chapter exercises and practice until you can do them fully on your own without notes, like in a test**
- 3. Write down your questions and bring them to class**